Indira Gandhi National Tribal University, Amarkantak, Madhya Pradesh, (484887)

Submitted to complete

Under Graduate Course (B.Voc.) in Software Development

Department of Vocational Education

in accordance with the provision of IGNTU Regulations

Year of Submission: 2024-27

End Semester Project

Sandipani School Web Portal

Full Stack Web Development (Frontend & Backend using HTML, CSS, JavaScript, PHP & MySQL)
(Course Code: SD-EIP-III-303)

Department of Vocational Education

Faculty of Technical Vocational Education And Skill Training

Submitted to:

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B.Voc – SoftwareDevelopment
Semester: III rd

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DECLARATION

We hereby declare that the project titled **"Sandipani School Web Portal"** is an original work completed as part of the **End Semester Project** for the **B.Voc. in Software Development** under the **Department of Vocational Education**, Faculty of Technical Vocational Education and Skill Training,
Indira Gandhi National Tribal University, Amarkantak (M.P.). We further declare that this project has not been submitted, wholly or partly, for any other course or examination.
All information, modules, design elements, and data presented in this report have been prepared based on our own work and guidance received from the faculty.

Project Members:
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B.Voc – Software Development
Semester: III

Date: _____
Place: Amarkantak (M.P.)

Signatures:

INDIRA GANDHI NATIONAL TRIBAL UNIVERSITY, AMARKANTAK
Department of Vocational Education
B.Voc. – Software Development

CERTIFICATE

This is to certify that the project titled **"Sandipani School Web Portal"** (Full Stack Web Development using HTML, CSS, JavaScript, PHP & MySQL) submitted by **Sachin lodhi(2401123025), Ajay Gupta (2401123003), Bhupendra mahra(2401123045),** in partial fulfillment of the requirements for the **Undergraduate Course (B.Voc.) in Software Development** during the **IIIrd Semester (2024-27)** has been carried out under my guidance.

The project work is **original** and represents the student's own efforts and findings.

Date:
Signature :

Submitted to:
Dr. Kamlesh Kumar Pandey
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Signature of Students:

ACKNOWLEDGEMENT

We feel immense pleasure in expressing our sincere gratitude to all those who have supported and guided us during the successful completion of our project titled **"Sandipani School Web Portal"**.

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We would also like to extend our heartfelt appreciation to the **senior batch students** of the department for their timely help, guidance, and sharing of knowledge, which greatly facilitated our project development.

Finally, we are grateful to our **family and friends** for their unwavering support and motivation, which encouraged us to complete this project successfully.

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Abstract

The project titled **"Sandipani School Web Portal"** aims to create a centralized departmental profile system to connect students, faculty, and alumni of the B.Voc SD program at IGNTU. Currently, the absence of a dedicated platform leads to scattered information, delayed notifications, and difficulties in managing academic and administrative records.

The portal addresses these challenges by providing a centralized platform for storing and managing student, alumni, and faculty information, offering real-time updates on departmental news, notices, events, and syllabus changes, and facilitating structured feedback collection. This ensures efficient communication, reduces administrative workload, and enhances the overall academic experience.

The system is developed using a **3-tier architecture**: HTML and CSS for frontend, PHP for server-side logic, and MySQL for data management. Key functionalities include semester-wise student profiles, faculty management, curriculum updates, notice and announcement systems, gallery management, alumni and placement records, and a secure admin panel with CRUD operations. Responsive design principles ensure accessibility across devices, while security measures such as session management and form validation protect data integrity.

By implementing this web portal, the SD department achieves improved coordination, streamlined academic management, and a more connected and interactive environment for both students and faculty.

INTRODUCTION

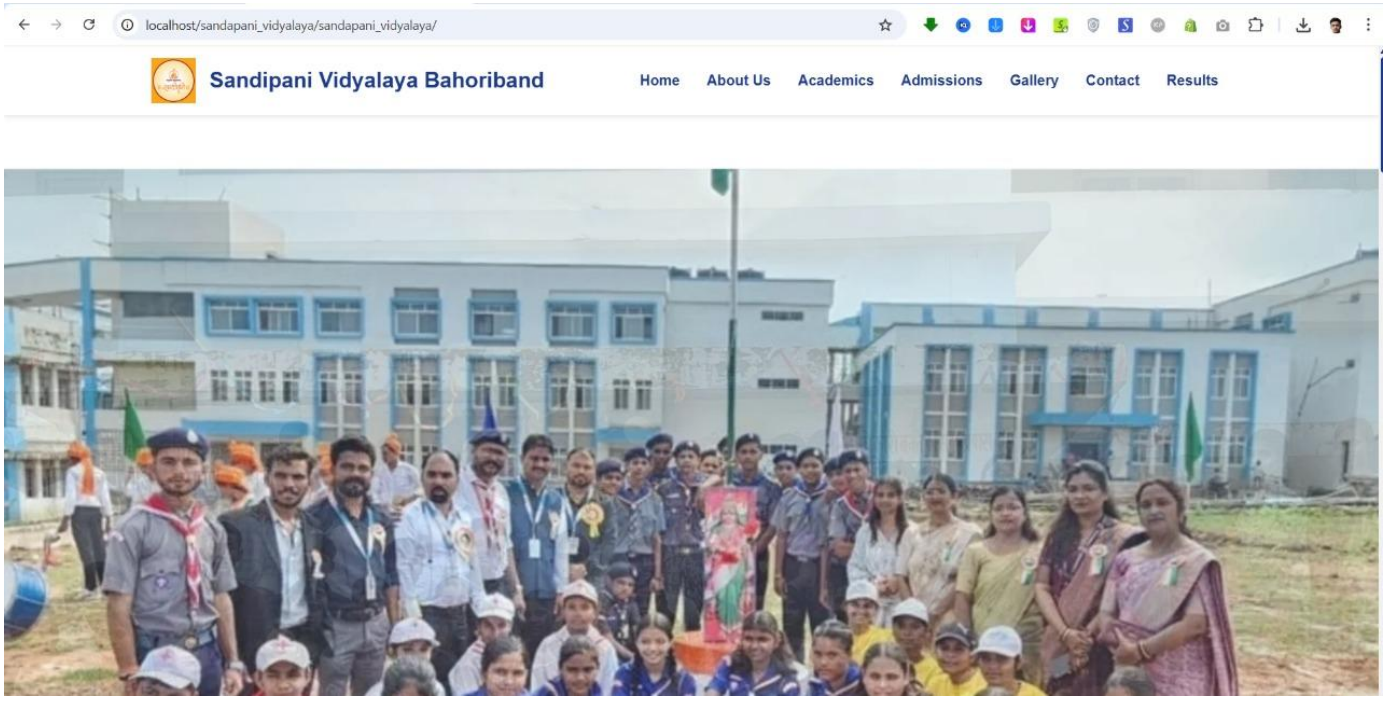
This project focuses on developing a comprehensive web portal for **Sandipani School, Bahoriband**. The primary objective is to create a centralized platform that serves as a common interface for the school and its community (students, parents, and prospective applicants), enabling efficient access to and management of academic and institutional information.

Currently, parents and students often face delays in receiving important updates regarding notices, school events, fee structures, and other activities. Similarly, the school administration requires an organized and modern method to communicate announcements and showcase the school's achievements. While traditional methods like physical notice boards or student diaries are used, they are often unreliable and fail to reach everyone in a timely manner. To address this communication gap, this project designs and develops a web portal exclusive to Sandipani School, ensuring precise and updated information is accessible to everyone at all times.

Key Features and Functionalities:

1. **Real-time Updates:** Provides regular information on school activities, events, and announcements.
2. **School Information:** Shares details about the school's history, vision, mission, and facilities.
3. **Academics Information:** Offers details on classes offered and a dynamic fee structure.
4. **Faculty & Student Management:** A backend system to manage teacher and student records.
5. **Digital Gallery:** Displays categorized photos of events and school life.
6. **Digital Notice Board:** Posts important deadlines, updates, and opportunities.
7. **Contact & Admissions:** Provides clear information and an online inquiry form.

The system is designed with a clean layout and intuitive navigation. By providing centralized data and real-time updates, this platform improves communication, streamlines administrative processes, and strengthens the connection between the school and its community. Overall, this project aims to create a more informed, organized, and interactive school environment, enhancing its brand identity and fostering better collaboration.



OBJECTIVES OF THE PROJECT

1. **To develop a centralized web portal for Sandipani School:**
   The system will consolidate all essential school information into a single, user-friendly interface, reducing confusion and improving accessibility for parents, students, and staff.

2. **To provide accurate and real-time school updates:**
   The platform will ensure the community receives timely notifications about events, notices, holidays, and deadlines, minimizing communication gaps.

3. **To maintain comprehensive records through an admin panel:**
   The backend system will allow administrators to manage profiles for students and teachers, facilitating organized data management and easy reference.

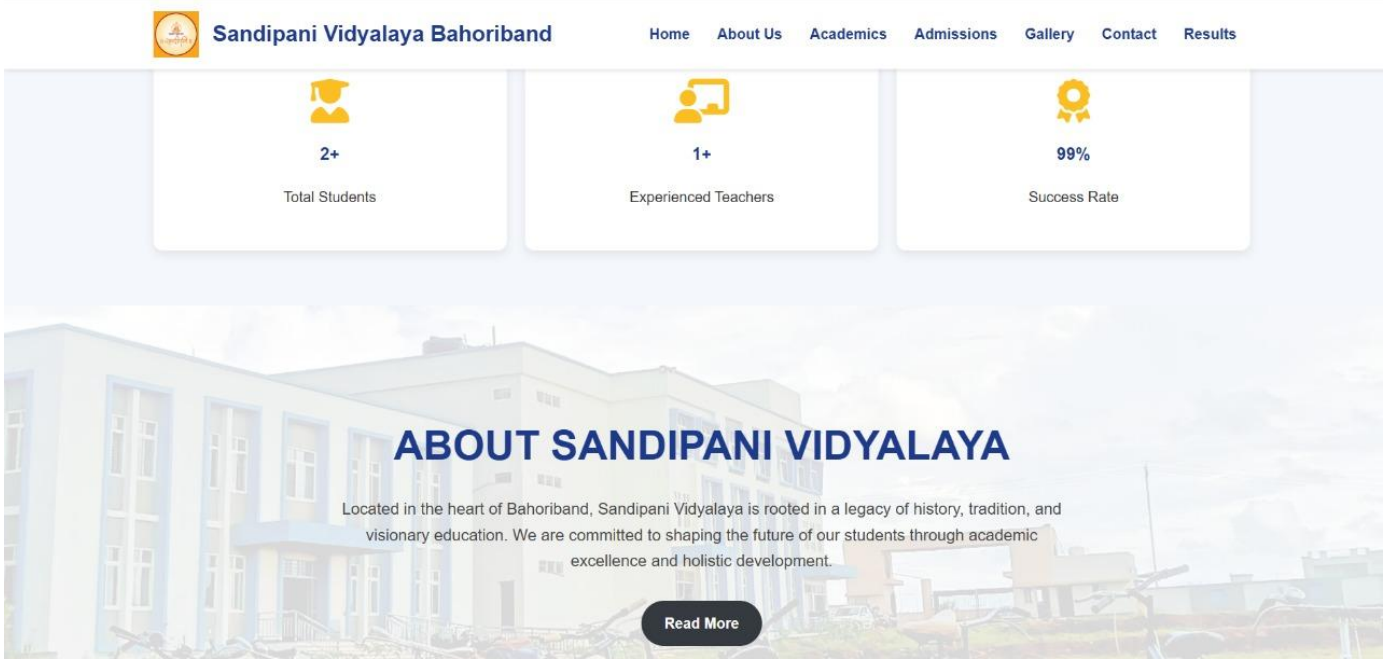
4. **To streamline the admission inquiry process:**
   By providing all necessary information online—such as admission procedures, required documents, and fee structures—the portal will reduce the number of repetitive queries handled by school staff.

5. **To create a digital gallery for school events:**
   Photos from school workshops, sports days, cultural events, and other activities will be organized and made accessible, preserving school memories and showcasing its vibrant culture.

6. **To improve communication between the school and parents:**
   The platform will serve as a reliable, 24/7 channel for sharing important announcements and updates, fostering a connected and interactive school environment.

7. **To ensure the system is secure and easy to manage:**
   A secure, password-protected admin panel with full CRUD capabilities will be developed, allowing non-technical staff to update the website's content without needing programming knowledge.

8. **To build a scalable and adaptable platform for future growth:**
   The system will be designed to support additional features in the future, such as online fee payments or a student result portal, ensuring long-term usability.



LIMITATIONS WITH RESPECT

Lack of a Centralized Information Platform: All information is scattered. Parents have to call or visit the school for basic details like fee structure or admission dates.

Inefficient and Delayed Communication: Important notices sent via student diaries may not reach parents on time or may be lost. There is no way to send urgent announcements to everyone instantly.

No Digital Showcase of School Activities: There is no official gallery to display photos of school events, achievements, or facilities, which is a missed opportunity for branding and community building.

High Administrative Workload: School staff spend a significant amount of time answering the same repetitive questions about admissions, fees, and schedules.

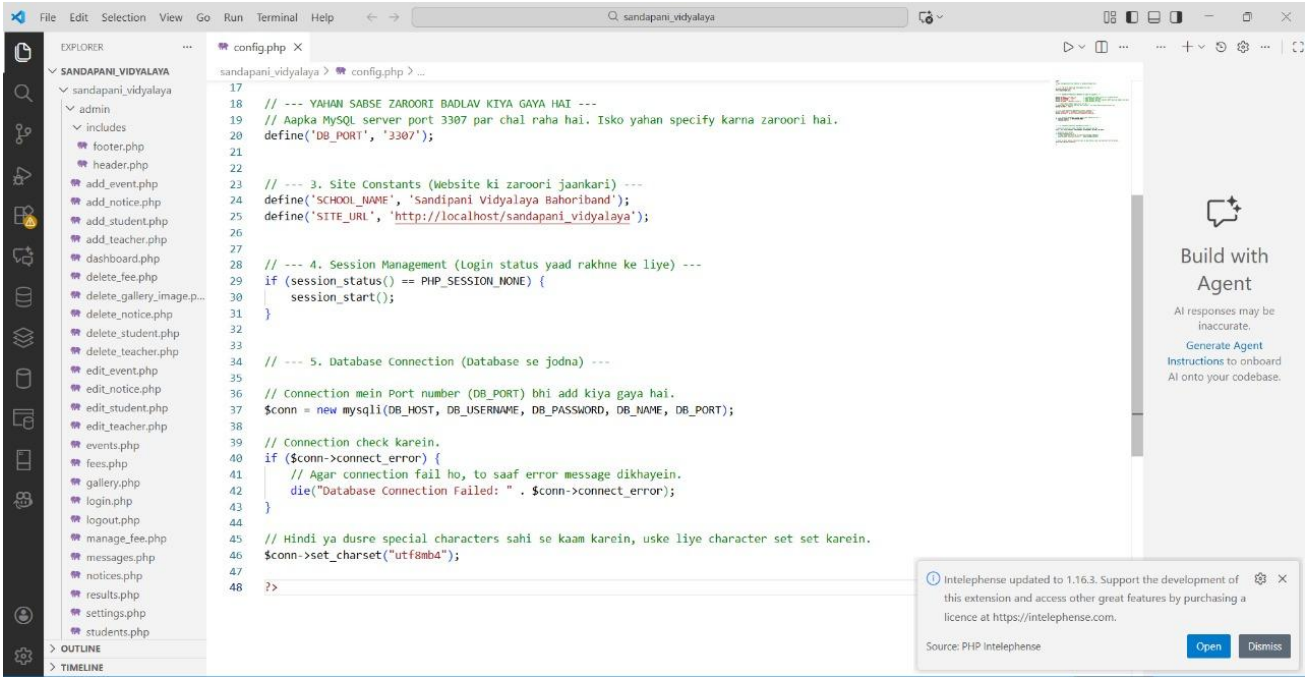
Absence of a Professional Digital Identity: Without a website, the school has a limited online presence, which can affect its perception among prospective, tech-savvy parents.

No System for Feedback: There is no formal channel for parents or students to provide suggestions or feedback to the school management.

Information is Not Accessible 24/7: Parents can only get information during school working hours.

SYSTEM ANALYSIS

- **Purpose and Scope of the Sandipani School Web Portal**
- The "Sandipani School Web Portal" is a web-based Information Management System (IMS) designed to centralize and streamline all communication, information sharing, and record management for the school.
- **System Architecture Overview**
- The system is designed using a **3-Tier Architecture**:
- **PRESENTATION LAYER (Frontend):** HTML, CSS, JavaScript
- **APPLICATION LAYER (Backend):** PHP Scripts, Validation, Processing
- **DATA LAYER (Database):** MySQL Database, Tables, Records
- **User Roles and Characteristics**
- **Admin / School Staff:**
- Has full system access through a secure login.
- Can perform CRUD operations (Add, Edit, Delete) on all dynamic content.
- Responsible for data entry and maintaining the website's accuracy.
- **Public Users (Parents, Students, Visitors):**
- Have read-only access to all public pages.
- Cannot modify any data or access the admin panel.
- Can submit inquiries through the contact form.
- **Functional Requirements**
- **FR-1: User Authentication System:** The system must provide a secure login for admin users.
- **FR-2: Notice Management:** Admin can create, view, update, and delete notices.
- **FR-3: Student & Teacher Management:** Admin can maintain records of students and teachers.
- **FR-4: Gallery Management:** Admin can upload images to different categories.
- **FR-5: Events Management:** Admin can add, edit, and delete school events.
- **FR-6: Fee Structure Management:** Admin can update the class-wise fee structure.
- **FR-7: Contact Message Viewing:** Admin can view all messages submitted via the contact form.



System Design

Architectural Design

The project follows a 3-Tier Architecture, which provides a clear separation of concerns:

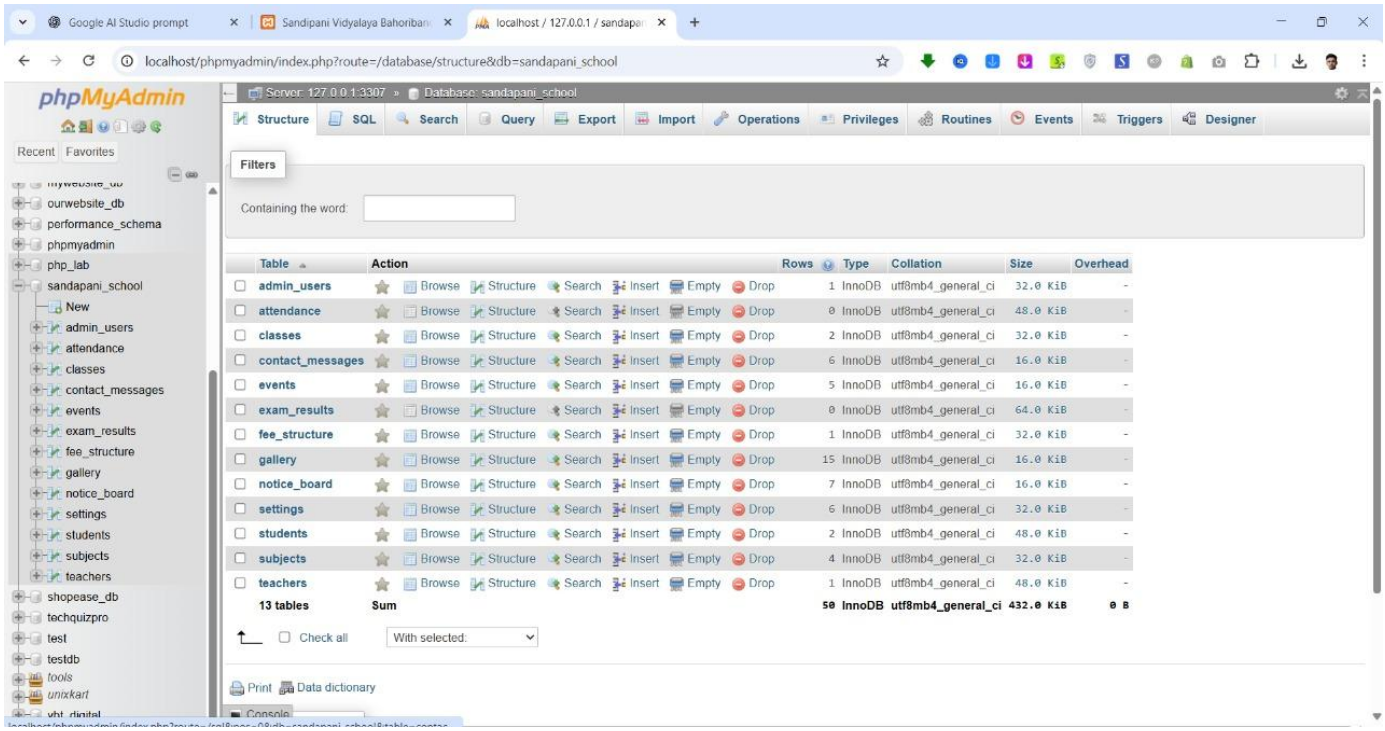
1. **Presentation Tier (Frontend):** This is what the user sees and interacts with. It consists of HTML pages styled with CSS and made interactive with JavaScript. It runs in the user's browser.
2. **Application Tier (Backend):** This is the "brain" of the system. It consists of PHP scripts running on the Apache server. It receives requests from the frontend, processes them (e.g., validates form data), communicates with the database, and generates the HTML to be sent back to the user.
3. **Data Tier (Database):** This is the storage unit. It consists of the MySQL database, which stores all the website's data in structured tables.

Database Design

The database, named sandapani_school, is designed with several normalized tables to ensure data integrity and avoid redundancy. The key tables are:

- admin_users: Stores hashed passwords and usernames for admin login.
- students, teachers: Store profiles of students and faculty.
- notices, events: Store time-sensitive information for the dynamic sections.
- gallery: Stores paths and categories for all uploaded images.
- fee_structure: Stores class-wise fee details.

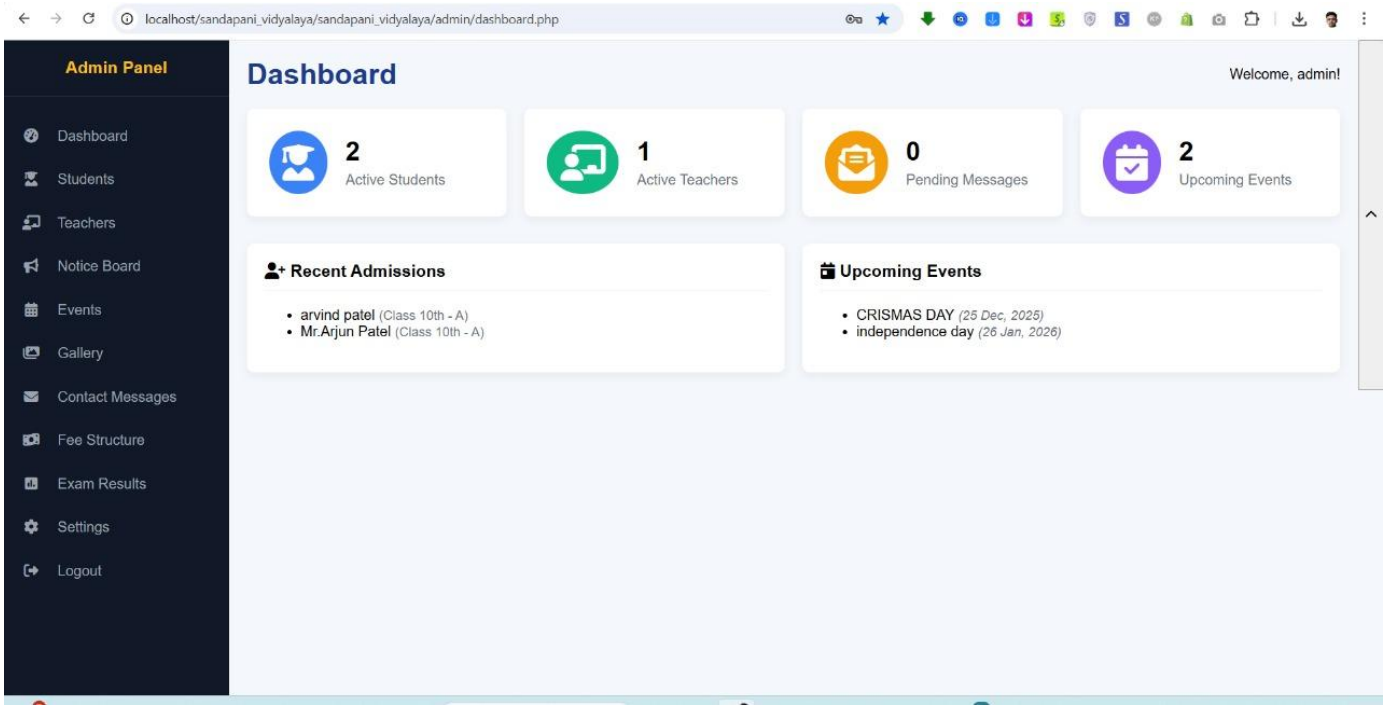
- contact_messages: Stores inquiries from the contact form.



Implementation

The implementation phase involved translating the system design into functional code.

1. **Environment Setup:** The project was developed on a local server environment using XAMPP, which provided the Apache server, MySQL database, and PHP interpreter.
2. **Frontend Development:** The user interface was built using HTML5 for structure, CSS3 for styling (including responsive design with media queries), and JavaScript for client-side functionalities like the image slider and form validation.
3. **Backend Development:** PHP was used as the server-side language.
    - A central config.php file was created to handle the database connection.
    - An includes folder was used to store reusable components like header.php and footer.php.
    - The **Admin Panel** was implemented with a secure login system using PHP sessions. CRUD functionalities were built for each module using HTML forms for input and PHP scripts to process the data and execute SQL queries (INSERT, UPDATE, SELECT, DELETE)



Testing & Quality Assurance

To ensure the website is robust and error-free, a multi-level testing strategy was employed.

1. **Unit Testing:** Individual components were tested in isolation. For example, the database connection function in config.php was tested independently.
2. **Integration Testing:** Different modules were tested together. For instance, a new notice was added via the admin panel, and the homepage was checked to see if it appeared correctly.
3. **System Testing:** The entire website was tested as a whole on different web browsers (Chrome, Firefox) and devices (desktop, mobile) to ensure cross-browser compatibility and responsive design.
4. **User Acceptance Testing (UAT):** The system was tested from the perspective of both a public user and an admin to ensure it is intuitive and meets all functional requirements.

Results & Discussion

The project has successfully resulted in the development of a fully functional and dynamic "Sandipani School Web Portal." The final output is a professional website that effectively addresses all the problems identified in the existing system.

- **The homepage now features a dynamic slider, real-time notice and event updates, and other key sections.**
- **The Admin Panel is fully operational, allowing for easy management of all website content.**
- **The communication gap has been bridged, as all information is now available 24/7 in a centralized location.**
- **The website successfully establishes a strong and modern digital identity for Sandipani School.**

Conclusion & Future Scope

Conclusion

The "Sandipani School Web Portal" project successfully achieves its objective of creating a comprehensive digital solution for the institution. By transitioning from inefficient manual processes to a centralized, dynamic, and user-friendly online platform, this project provides a robust foundation for the school's digital transformation. The development of a clear public-facing website coupled with a secure Admin Panel empowers the school to manage its digital presence effectively, creating a more connected, informed, and engaged community.

Future Scope

The scalable architecture of this project allows for significant future enhancements. Potential additions to evolve this portal into a full-fledged School ERP system include:

- **Student/Parent Login Portals:** For accessing personalized information like academic performance and attendance.
- **Online Fee Payment Gateway:** To automate the fee collection process.
- **Digital Result Management System:** For students to view their report cards online.
- **Alumni Network Portal:** To foster a strong and engaged alumni community.

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